

EPILEPSY DIGITAL TWIN REPORT • INTEGRATED NEUROLOGICAL DECISION SUPPORT

Neuro Digital Twin Integrated Clinical Report

An integrated clinical assessment combining follow-up visits, seizure history, EEG, ECG, CT/MRI imaging, laboratory results, and pharmacogenomic testing for Focal impaired-awareness seizures.

Patient & Clinical Context		10/10 data categories included · generated 2026-09-11T16:57:32				
PATIENT John Doe Patient ID SYN-100482	AGE / SEX 34 years · Male	PRIMARY DIAGNOSIS Focal impaired-awareness seizures Secondary diagnosis: Secondary generalized tonic-clonic seizures	DIAGNOSED SINCE 12 May 2021			
CURRENT CLINICAL STATUS Stable outpatient epilepsy state with persistent breakthrough seizure risk						
CURRENT THERAPY Levetiracetam (Keppra) 1000 mg BID · Lamotrigine 150 mg BID						
REPORTING PERIOD Data reviewed: 01 Jan – 30 Jun 2026 (~6 months) Clinic visit notes cover 01 Jan 2026 to 02 Mar 2026; laboratory, imaging, EEG/ECG, and questionnaire data continue through 30 Jun 2026.						
5 Clinical Visits	12 CBC Reports	6 CT Studies	6 MRI Series	12 ECG Sessions	35 EEG Recordings	12 Questionnaires

01 Executive Summary

Across five clinical-note visits from 2026-01-01 to 2026-03-02, the patient has chronic focal epilepsy with persistent breakthrough seizures on levetiracetam and lamotrigine. The clinician repeatedly documented reasonably stable seizure control and favorable quality-of-life trajectory, while seizure burden, sleep quality, trigger exposure, aura symptoms, and medication tolerability continued to fluctuate.

67% Overall Health Score	Moderate Overall Clinical Risk	Elevated EEG-Derived Seizure Risk	10 (7 high) Genetic Safety Flags
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Domain the patient rates worst on questionnaires: Epileptic Triggers. See the Digital Twin Risk Dashboard (Section 13) for the full risk breakdown.

02 6-Month Timeline

- 01 JAN 2026

Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.
- 16 JAN 2026

Seizure clustering documented during follow-up — Raises near-term seizure-monitoring priority despite otherwise stable clinician impression.
- 03 FEB 2026

CBC/LFT safety review documented — Supports medication-safety monitoring without a documented abnormal lab signal.
- 15 FEB 2026

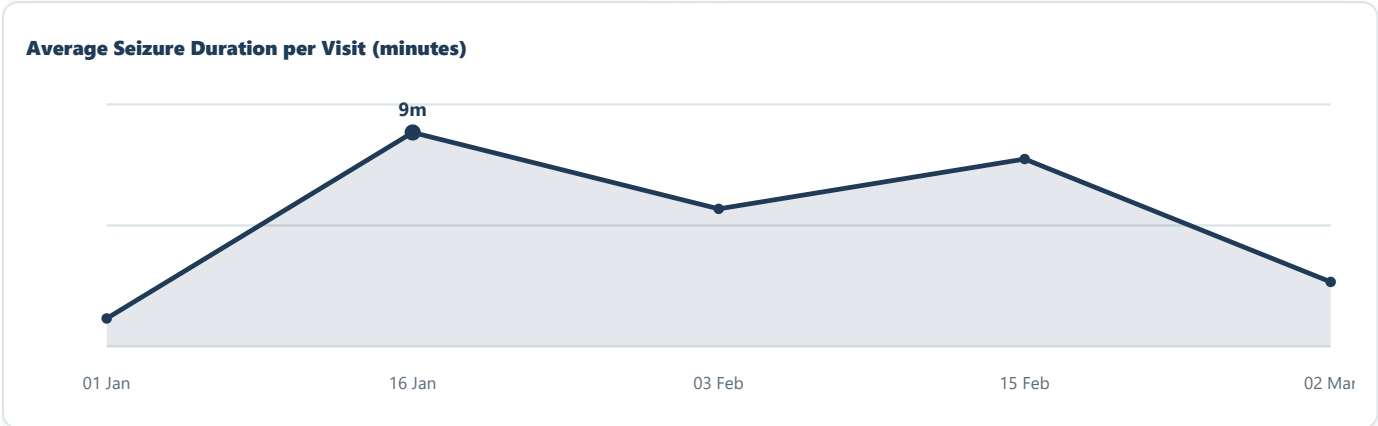
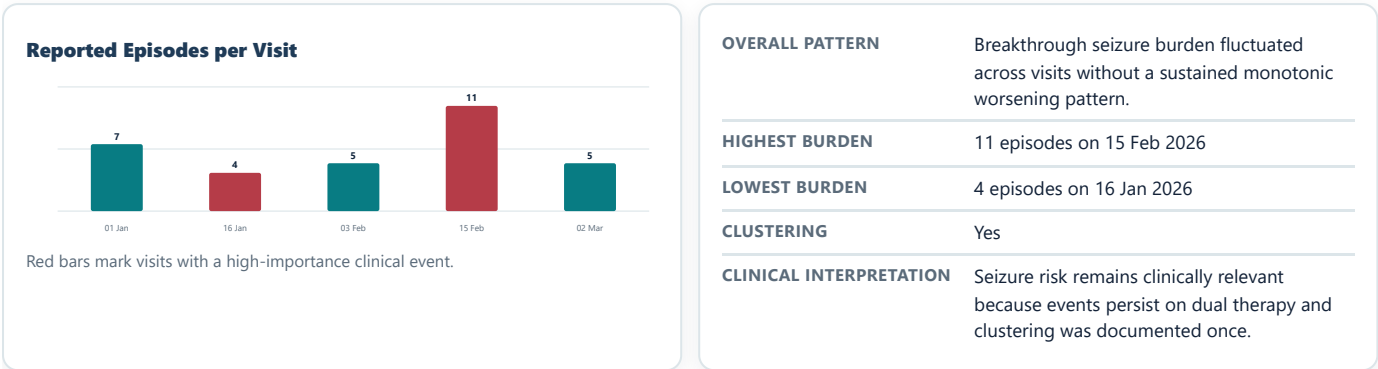
Highest recorded seizure burden in clinical-note window — 11 reported episodes make this the peak burden visit.
- 02 MAR 2026

Routine epilepsy follow-up with stable clinician impression — Contributes to longitudinal assessment of seizure burden, triggers, sleep, and tolerability.
- 10 SEP 2026

Pharmacogenomic (PGx) panel reported — 76 variants analyzed across 47 medications; see Pharmacogenomics.

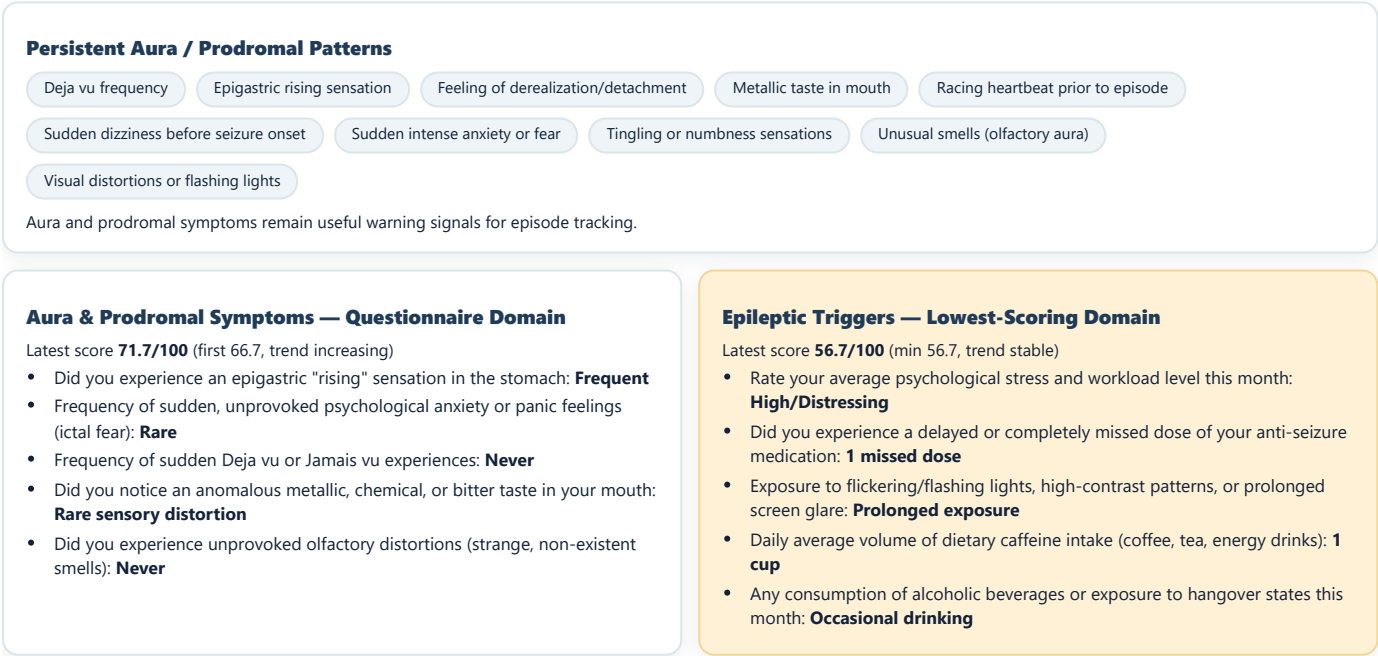
03 Seizure Trends

Clinical-note episode counts, clustering, and duration



04 Aura & Triggers

Prodromal patterns and patient-reported trigger exposure



05 Medication Response

Current regimen, tolerability, and reported side effects

Current Regimen

Drug	Dose
Levetiracetam (Keppra)	1000 mg BID
Lamotrigine	150 mg BID

Effectiveness	Partial response
Tolerability	Generally tolerated with intermittent dizziness and variable side-effect burden
Documented Toxicity	No
Clinical Interpretation	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Persistent Side Effects (clinical notes)

Memory fog post-dose

Tremor in hands

Weight change since last visit

Gum hypertrophy

Mood irritability

Nausea after taking medication

Skin rash or itching

Daytime drowsiness

Dizziness/lightheadedness

Unsteady gait (ataxia)

Medication Side Effects — Questionnaire Domain

Latest score **78.3/100** (first 68.3, trend increasing)

- Rate the severity of daytime drowsiness or somnolence: **Mild (Manageable)**
- Experience of dizziness, vertigo, or lightheadedness after taking medication: **Never**
- Presence of physical instability, unsteadiness, or clumsy movements (ataxia): **Slight imbalance**
- Frequency of unprovoked irritability, anger, or mood volatility: **Mild irritability**
- Signs of dermatological distress or unexplained skin rashes: **Clear skin**

06 Pharmacogenomics

76 variants across 47 medications · reported 10 Sep 2026

Pharmacogenomic Panel Summary

CYP2D6 GG	CYP2C19 GA	CYP2C9 *1/*3 - Intermediate Metabolizer	CYP2B6 *1/*6 - Intermediate Metabolizer	CYP3A5 *3/*3 - Poor Metabolizer
CYP1A2 AA	CYP3A4 TT	UGT1A4 *1/*3A - Decreased Function	POR *1/*28 - Decreased Function	

Interpretive method: a pharmacogenomic association is weighed against this patient's own observed clinical response, laboratory, and EEG evidence -- it is never used as an isolated prescribing instruction, and a favorable genotype cannot override documented toxicity or persistent symptoms. The table below shows the single most clinically significant finding per current-regimen drug; the full therapy-level assessment -- this finding combined with clinical, EEG, laboratory, and external-database evidence -- is presented in Section 07, Drug Interactions.

DRUG	GENE (GENOTYPE)	WORST / MOST SIGNIFICANT OUTCOME	CITATION
Lamotrigine	CYP3A4 (TT)	REDUCED EFFICACY	PMID:28343093
Levetiracetam	SCN1A (CT)	EFFICACY	PMID:28753467

External Database Evidence — PubMed / ClinVar / CPIC

Database evidence only, not patient-verified: each row below is a live public-database lookup for the same gene/drug pair as the patient-specific finding in the table above -- it establishes general scientific plausibility for that pair, not a fact confirmed for this individual patient. An UNRESOLVED result means this exact query returned nothing, or failed, at the time checked -- it is not evidence that no relevant data exists, and must not be read as a clean or reassuring negative.

GENE → DRUG (PATIENT-SPECIFIC PAIR)	CPIC	CLINVAR	PUBMED
CYP3A4 → Lamotrigine	UNRESOLVED CPIC lists 'lamotrigine' (RxNorm:28439) but has no CYP3A4-lamotrigine pair on record.	VCV004885026: UNCERTAIN SIGNIFICANCE 70 total ClinVar record(s) for this gene (gene-wide count, not variant-specific) Accession VCV004885026 · Retrieved 11 Sep 2026	PMID 21635243: Effects of lamotrigine and phenytoin on the pharmacokinetics of atorvastatin in healthy volunteers. PMID 21635243 · Retrieved 11 Sep 2026
Note: top table = citation from patient's own PGx report; table below = independent live PubMed search, may differ.			
SCN1A → Levetiracetam	UNRESOLVED CPIC has no drug entry matching 'levetiracetam'; no gene-drug pair can exist for it.	VCV004887952: UNCERTAIN SIGNIFICANCE 5406 total ClinVar record(s) for this gene (gene-wide count, not variant-specific) Accession VCV004887952 · Retrieved 11 Sep 2026	PMID 39200163: The Biallelic Inheritance of Two Novel SCN1A Variants Results in Developmental and Epileptic Encephalopathy Responsive to Levetiracetam. PMID 39200163 · Retrieved 11 Sep 2026
Note: top table = citation from patient's own PGx report; table below = independent live PubMed search, may differ.			

Evidence Scope: live-verified via ClinVar, CPIC, DailyMed, PubMed, and ClinicalTrials.gov only. The rest of the NeuroTwin resource catalog (PharmGKB, DrugBank, Reactome, STRING, GTEx, and others) is out of scope for this live-lookup layer and is not reflected anywhere in this table — see Known Limitations below.

Known Limitations (External Evidence):

- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
- A 'no_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
- Responses may be served from the on-disk cache in reports/external_evidence/.cache/ rather than a fresh call; each record's 'from_cache' and 'retrieved_at' fields state which. Pass --refresh to force live calls.

07 Drug Interactions

2 current medication(s) reconciled · Flockhart Cytochrome P450 Drug Interaction Table (curated-snapshot-2024)

DDI & Patient-Specific Therapy Assessment — Summary

Overall: 1 clinically relevant findings | 2 unresolved finding(s).

2

Current Medications Screened

2

Resolved in Curated DDI Source

0

Unresolved / Not in Curated Source

0

Not Evaluated

DRUG PAIR	FLAG	KEY FINDING	PATIENT IMPACT	EVIDENCE SOURCE
Lamotrigine + Levetiracetam (Keppra)	NO MAJOR CONCERN	No resolved major DDI identified between these two current medications.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression. This reflects therapy-level effectiveness, not a confirmed drug interaction.	Curated DDI (Flockhart); External evidence (PubMed); Patient pharmacogenomic findings
Lamotrigine	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings
Levetiracetam (Keppra)	UNRESOLVED	Therapeutic evidence present, but clinical effectiveness remains incomplete.	Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.	Patient pharmacogenomic findings; Laboratory findings; Clinical record; EEG findings

Key clinical interpretation:

These flags identify findings that require ongoing clinical monitoring, not an automatic medication change. Each finding should be interpreted alongside the patient's actual treatment response, adverse effects, and the clinical/laboratory findings documented elsewhere in this report -- a resolved or unresolved flag is not itself a prescribing instruction.

Monitoring / Clinician Review Checklist

Medication reconciliation (dose, frequency, and route consistent across all source records)

Actual treatment response (seizure/symptom control on the current regimen)

Relevant adverse effects reported since the last review

Therapeutic drug levels, when indicated and available

Relevant laboratory and clinical parameters (e.g. renal, hepatic, hematological)

Medication Reconciliation

DRUG	DOSE	FLAGS
Lamotrigine	150.0 mg BID	—
Levetiracetam (Keppra)	1000.0 mg BID	—

No conflicting doses found for the same drug across the reconciled source data.

Current Regimen — Curated-Source Coverage

MEDICINE	STATUS	MEANING	ACTION
Lamotrigine	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	No additional action required beyond the current-current pair evidence below.
Levetiracetam (Keppra)	RESOLVED	Curated CYP/pharmacokinetic reference data exists for this drug (Flockhart Cytochrome P450 Drug Interaction Table).	No additional action required beyond the current-current pair evidence below.

Clinical inference:

Absence of a resolved current-current interaction does not establish absence of interaction. Unresolved predicted-model coverage limits interpretation; the active medication list should be reassessed against a complete interaction source before any treatment change.

SuperCYPsPred predicted pharmacokinetic evidence is not available locally for any drug in this regimen.

No therapeutic drug concentration data is present in the source reports.

No renal function (eGFR) data is present in the source reports.

Therapy-Level Evidence Assessment:

for each current medication, the supporting, counter, and unresolved evidence identified across pharmacokinetic, pharmacodynamic, pharmacogenomic, clinical, EEG, and laboratory sources -- not a pass/fail interaction check. Absence of data does not establish absence of risk; "unresolved" is never presented as "no interaction." Recommended monitoring appears per therapy below (see also the section-level checklist above).

Generated from validated patient data. Decision-support summary only -- not a substitute for clinician review.

Page 5 of 11

Lamotrigine

EFFECTIVENESS INCOMPLETE

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- SCN1A (CT): efficacy — findings_by_therapeutic_class
- CACNA1H (AG): efficacy — findings_by_therapeutic_class
- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response

Counter-Evidence

- ABCB1 (GG): reduced efficacy — findings_by_therapeutic_class
- CYP3A4 (TT): reduced efficacy — findings_by_therapeutic_class
- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- Seizure clustering was documented during the observation window. — clinical_inference.seizure_analysis
- EEG synthesis remained consistent with active epilepsy (EEG-derived risk: Elevated; future seizure probability: Elevated). — digital_twin_state

Unresolved Evidence

- No renal function (eGFR), hepatic enzyme, or therapeutic drug concentration data is present in the available source reports for this patient. — not available in source data
- PubMed candidate reference (PMID 28862044): "Acute lamotrigine overdose: a systematic review of published adult and pediatric cases." -- requires study-level review before use as a patient-specific finding. — pubmed.ncbi.nlm.nih.gov/28862044/
- DailyMed label on file: "LAMOTRIGINE TABLET, FILM COATED, EXTENDED RELEASE [AMNEAL PHARMACEUTICALS LLC]" (published Aug 20, 2026); label section text (warnings/interactions/contraindications) was not extracted by this lookup and requires direct review. — dailymed.nlm.nih.gov...d=0833d3df-2df1-4abd-abe0-7fb2c0253c80
- ClinicalTrials.gov: no trial matched this exact condition/drug query at the time checked. — ClinicalTrials.gov (API v2)
- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no SCN1A-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004887952: Uncertain significance (5406 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4887952/
- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no ABCB1-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004858346: Uncertain significance (797 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4858346/
- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no CYP3A4-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004885026: Uncertain significance (70 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4885026/
- CPIC: CPIC lists 'lamotrigine' (RxNorm:28439) but has no CACNA1H-lamotrigine pair on record. — CPIC (api.cpicpgx.org)
- ClinVar VCV004886626: Uncertain significance (4289 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4886626/
- Lamotrigine was not resolved against a live drug-drug interaction knowledge base in the available sources; only curated reference pharmacology and this patient's own pharmacogenomic findings were evaluated. — curated reference pharmacology only (see reference_data)

Clinical Impression

4 supporting, 5 counter, and 13 unresolved evidence item(s) were identified for Lamotrigine. Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

- Continue EEG-derived seizure-risk monitoring.
- Continue periodic CBC and safety-laboratory monitoring.
- Continue seizure frequency, duration, and clustering tracking.
- Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

Levetiracetam (Keppra) EFFECTIVENESS INCOMPLETE

Therapeutic evidence present, but clinical effectiveness remains incomplete.

Supporting Evidence

- SCN1A (CT): efficacy — findings_by_therapeutic_class
- CBC status across the observation window: stable within reference ranges; no documented hematological toxicity. — digital_twin_state
- No documented drug toxicity for this agent in the clinical-notes source. — clinical_inference.medication_response

Counter-Evidence

- Breakthrough seizures persisted on the current regimen, with a recorded peak of 11 episodes on 2026-02-15. — clinical_inference.seizure_analysis
- Seizure clustering was documented during the observation window. — clinical_inference.seizure_analysis
- EEG synthesis remained consistent with active epilepsy (EEG-derived risk: Elevated; future seizure probability: Elevated). — digital_twin_state

Unresolved Evidence

- No renal function (eGFR), hepatic enzyme, or therapeutic drug concentration data is present in the available source reports for this patient. — not available in source data
- PubMed candidate reference (PMID 30036676): "Levetiracetam." -- requires study-level review before use as a patient-specific finding. — pubmed.ncbi.nlm.nih.gov/30036676/
- DailyMed label on file: "LEVETIRACETAM TABLET [REMEDYREPACK INC.]" (published Sep 07, 2026); label section text (warnings/interactions/contraindications) was not extracted by this lookup and requires direct review. — dailymed.nlm.nih.gov...d=2d307850-0ed5-4f32-ac62-0e1e2fe27a7a
- ClinicalTrials.gov NCT00537238 (COMPLETED): "Pregabalin Versus Levetiracetam In Partial Seizures" -- registration does not establish study quality or results. — clinicaltrials.gov/study/NCT00537238
- CPIC: CPIC has no drug entry matching 'levetiracetam'; no gene-drug pair can exist for it. — CPIC (api.cpicgx.org)
- ClinVar VCV004887952: Uncertain significance (5406 total gene-wide record(s) for this gene, not variant-specific). — www.ncbi.nlm.nih.gov/clinvar/variation/4887952/
- Levetiracetam (Keppra) was not resolved against a live drug-drug interaction knowledge base in the available sources; only curated reference pharmacology and this patient's own pharmacogenomic findings were evaluated. — curated reference pharmacology only (see reference_data)

Clinical Impression

3 supporting, 3 counter, and 7 unresolved evidence item(s) were identified for Levetiracetam (Keppra). Clinical-notes inference: Current antiseizure therapy is associated with stable clinician impressions but incomplete seizure suppression.

Recommended Monitoring

- Continue EEG-derived seizure-risk monitoring.
- Continue periodic CBC and safety-laboratory monitoring.
- Continue seizure frequency, duration, and clustering tracking.
- Review the priority safety flags in the Pharmacogenomics section before adding or changing an interacting medication.

External Database Evidence — DailyMed / ClinicalTrials.gov / PubMed

Database evidence only, not patient-verified: each row below is a live public-database lookup keyed on the drug name from the patient's own current-regimen medication reconciliation above -- it establishes general plausibility for that drug, not a fact confirmed for this individual patient's actual dose or response. An UNRESOLVED result means this exact query returned nothing, or failed, at the time checked -- it is not evidence that no relevant data exists, and must not be read as a clean or reassuring negative.

DRUG (FROM PATIENT'S CURRENT REGIMEN)	DAILYMED LABEL	CLINICALTRIALS.GOV	PUBMED
Lamotrigine	LAMOTRIGINE TABLET, FILM COATED, EXTENDED RELEASE [AMNEAL PHARMACEUTICALS LLC] Published Aug 20, 2026 SetID 0833d3df-2df1-4abd-abe0-7fb2c0253c80 · v20 · Retrieved 11 Sep 2026	UNRESOLVED No trial matched this exact condition/drug query at the time checked.	PMID 28862044: Acute lamotrigine overdose: a systematic review of published adult and pediatric cases. PMID 28862044 · Retrieved 11 Sep 2026
Levetiracetam	LEVETIRACETAM TABLET [REMEDYREPACK INC.] Published Sep 07, 2026 SetID 2d307850-0ed5-4f32-ac62-0e1e2fe27a7a · v4 · Retrieved 11 Sep 2026	NCT00537238: Pregabalin Versus Levetiracetam In Partial Seizures COMPLETED · 2 total match(es) NCT NCT00537238 · Retrieved 11 Sep 2026	PMID 30036676: Levetiracetam. PMID 30036676 · Retrieved 11 Sep 2026

Evidence Scope: live-verified via ClinVar, CPIC, DailyMed, PubMed, and ClinicalTrials.gov only. The rest of the NeuroTwin resource catalog (PharmGKB, DrugBank, Reactome, STRING, GTEx, and others) is out of scope for this live-lookup layer and is not reflected anywhere in this table — see Known Limitations below.

- Known Limitations (External Evidence):
- This module queries exactly five public APIs (PubMed, ClinVar, DailyMed, CPIC, ClinicalTrials.gov); it does not cover the rest of the NeuroTwin resource catalog (Reactome, STRING, GTEx, RCSB PDB, etc.) -- those remain out of scope for this module and must not be inferred from it.
 - A 'no_results' status means the API returned zero matching records for the exact query issued; it is not evidence of absence, and must not be reported as a negative finding.
 - Responses may be served from the on-disk cache in reports/external_evidence/.cache/ rather than a fresh call; each record's 'from_cache' and 'retrieved_at' fields state which. Pass --refresh to force live calls.

Limitations of this screen: This assessment is clinical decision support, not an automated prescribing system; no dose, start, or stop decision is made or implied here. No proposed/candidate medication list is present in the source data, so only the current regimen's pairwise and therapy-level evidence is evaluated. Curated pharmacokinetic

(CYP) coverage is limited to a small bundled reference table; a drug not listed there is reported as unresolved, not as free of interactions. Therapy assessments also draw on the live external-evidence layer (see Known Limitations under External Database Evidence, below, for its source-by-source scope and caveats).

08EEG

Longitudinal interictal / preictal / ictal signal characteristics

35
Total Recordings

12
Interictal

11
Preictal

12
Ictal

Signal Variance Midpoint by Phase (µV²)

Phase	Signal Variance Midpoint (µV²)
Interictal	344.45
Preictal	4343.59
Ictal	42395.2

EEG-derived risk: Elevated

Future seizure probability: Elevated

Dominant feature: Signal Variance

Phase Analysis

PHASE	STATE	ACTIVITY	VARIABILITY
Interictal	Baseline	Stable	Low
Preictal	Transition	Increasing Instability	Moderate-High
Ictal	Seizure	Hypersynchronous	Very High

PHASE	AMPLITUDE RANGE	STD DEV RANGE	VARIANCE RANGE
Interictal	-89; 99	11.68; 23.5	136.44; 552.46
Preictal	-345; 440	14.48; 92.07	209.71; 8477.48
Ictal	-504; 784	52.5; 286.42	2756.47; 82033.96

Longitudinal EEG recordings demonstrate a consistent transition from stable interictal activity to preictal electrical instability followed by high-amplitude ictal seizure activity. Findings are consistent with the dataset's interictal/preictal/ictal phase taxonomy, indicating reproducible seizure-related signal evolution; this is a signal-classification observation, not a clinician-verified diagnosis.

09MRI / CT

Structural imaging: neuroimaging key findings

CT — 6 studies, 12 Jan – 29 Jun 2026

DATE	IMPRESSION
12 Jan 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
16 Feb 2026	No gross acute abnormality is evident on this limited skull-base level image; assessment is limited by slice level and image quality.
11 Mar 2026	No acute intracranial abnormality is evident on this single-image model-assisted review.
20 Apr 2026	No acute intracranial abnormality is evident on this high-convexity image; ventricular assessment is not available at this slice level.
14 May 2026	No acute intracranial abnormality is evident on this limited-quality near-vertex image.
29 Jun 2026	No acute intracranial abnormality is evident. Incidental bilateral choroid plexus calcifications are present.

Across the available CT image files, no gross acute intracranial hemorrhage, midline shift, or mass effect is evident on this model-assisted image-only review. Direct progression assessment is limited because the files represent single slices with different slice levels, resolutions, and image quality. This output is not a radiologist-verified diagnostic report.

MRI — Metadata-supported structural stability, diagnostic interpretation not available

FINDING	STATUS	DETAIL
Brain Morphology	NOT DIRECTLY ASSESSED	The provided files include volumetric MRI metadata, but no radiologist interpretation or morphometric segmentation was available for direct morphology assessment.
Brain Volume	STABLE WITHIN AVAILABLE METADATA	T1w nonzero volume range: 8387642 to 8777268 mm3 (mean 8641065 mm3; range 4.5%). FLAIR nonzero volume range: 6022385 to 6625690 mm3 (mean 6231344 mm3; range 9.7%). These are preprocessing-derived nonzero voxel volumes, not formal brain volumetry.
Tissue Signal Characteristics	QUANTITATIVELY SUMMARIZED, NOT DIAGNOSTICALLY INTERPRETED	FLAIR mean intensity 28.76 to 32.27; standard deviation 47.57 to 54.16; T1w mean intensity 99.82 to 105.56; standard deviation 153.13 to 162.05. Intensity statistics are useful for preprocessing consistency but are not diagnostic tissue-signal interpretation.

Not directly assessed from the available imaging metadata (this means "not evaluated," not "abnormal"): White Matter, Hippocampus, Cortical Structure, Mass Lesion, Hemorrhage, Infarction.

Stable by available preprocessing metadata. 6 MRI series were available across 3 study identifiers, including 3D T1-weighted MRI, T2-FLAIR MRI. Preprocessed metadata show consistent 3D acquisition characteristics and no metadata-level evidence of major global structural progression. A radiology report, lesion segmentation, hippocampal volumetry, and cortical thickness analysis were not available, so subtle pathology cannot be excluded from these metadata alone.

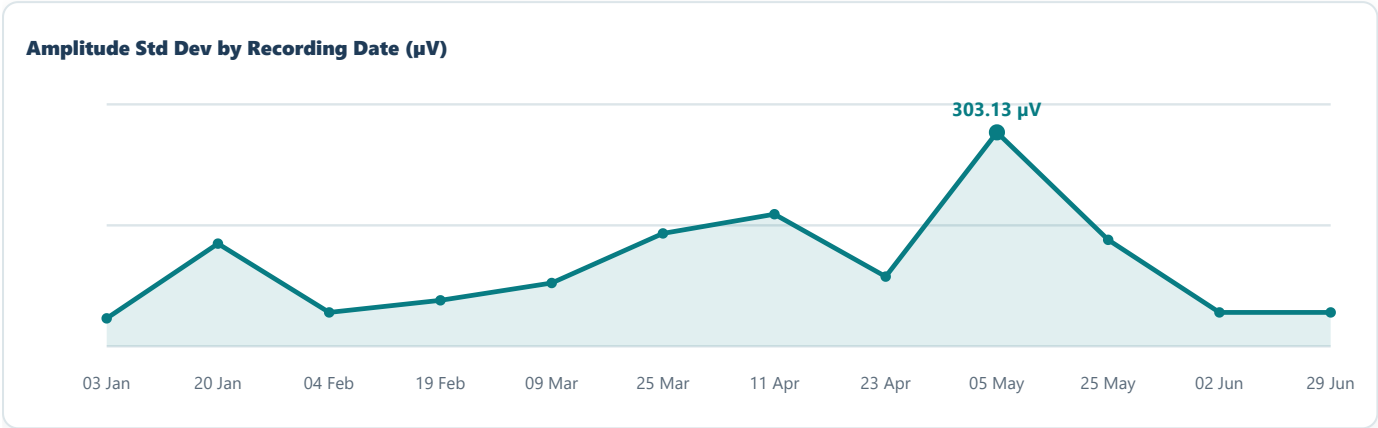
10ECG

single channel ("ECG SD") in every recording

12
Recordings

99h 11m
Total Duration

256
Sampling Rate (Hz)



- All 12 recordings use a single channel sampled at 256 Hz, so statistics are directly comparable across sessions.
 - Recording duration varies considerably, from about 1h 11m (2026-04-11) to about 22h 27m (2026-03-25).
 - The 2026-06-02 and 2026-06-29 recordings have matching percentile values and near-identical histogram shapes, suggesting very similar signal characteristics or a data duplication between the two sessions.
 - Standard deviation of amplitude ranges from about 82.6 uV (2026-01-03) to about 303.1 uV (2026-05-05), indicating substantially different signal variability/noise levels between sessions.
 - Peak-to-peak amplitude (max minus min) ranges from roughly 2,483 uV (2026-05-25) to about 3,577 uV (2026-03-25).
- This report is derived strictly from the statistical fields present in the source dataset. It does not represent a clinical or diagnostic ECG interpretation. Any assessment of cardiac rhythm, heart rate, or abnormality requires review of the underlying waveform by a qualified clinician or an appropriately validated diagnostic algorithm, neither of which is reflected in this report.

11CBC / Labs

12 reports, 01 Jan – 30 Jun 2026

Overall hematological status: Stable within reference ranges. No CBC parameter abnormality was identified against the report-provided reference ranges across the observation period.

PARAMETER	FIRST	LATEST	RANGE	TREND
Hemoglobin (Hb)	14.4	15.1	13.5 – 15.9	increasing
Total RBC count	5.0	5.1	4.8 – 5.3	increasing
Packed Cell Volume (PCV)	44.7	43.5	42.2 – 48.4	decreasing
Mean Corpuscular Volume (MCV)	90.0	97.8	88.4 – 98.8	increasing
MCH	27.6	29.9	27.6 – 31.1	increasing
MCHC	33.1	33.0	32.9 – 34.3	decreasing
RDW	12.1	13.5	12.1 – 13.5	increasing
Total WBC count	5761.0	9630.0	5476.0 – 9973.0	increasing
Neutrophils	58.0	54.0	52.0 – 61.0	decreasing
Lymphocytes	22.0	29.0	22.0 – 34.0	increasing
Eosinophils	4.0	4.0	1.0 – 5.0	stable
Monocytes	3.0	3.0	1.0 – 8.0	stable
Basophils	1.0	1.0	0.0 – 2.0	stable
Platelet Count	251725.0	209427.0	189551.0 – 373102.0	decreasing

Patient-reported vs. measured discrepancy: the 30 Jun 2026 questionnaire records patient recollections of White Blood Cell (WBC) count baseline trend from your latest Complete Blood Count (CBC): Mild Leukopenia (Borderline low); Platelet Count profile stability from recent blood evaluations: Borderline Thrombocytopenia (Low); Liver Function Test (LFT) - SGOT/AST status: Mild Elevation (40-100 U/L); measured CBC values across all 12 reports in this period remained within their reference ranges throughout (see table below). The discrepancy likely reflects patient recall/perception rather than a measured lab abnormality.

12

Sleep, Cognition & Quality of Life

Sleep		Cognition		Quality of Life	
TREND	Sleep remained variable across visits	MEMORY	Stable on clinician review, with low to intermittent patient-reported memory difficulty	DAILY FUNCTION	Independence in daily activities was repeatedly normal, intact, or stable
AVERAGE	6.2 hours per night across printed assessments	ATTENTION	Variable patient-reported concentration and attention symptoms	SOCIAL FUNCTION	Intermittent social withdrawal or isolation symptoms were reported
Daytime naps needed		LANGUAGE	Mild word-finding difficulty was noted but not described as worsening	CLINICAL INTERPRETATION	
Difficulty falling back asleep		MENTAL CLARITY	Improved mental clarity was reported at the latest visit	Quality of life remains affected by seizures, sleep, triggers, and medication tolerability, despite generally stable outpatient status.	
Fatigue level through the day					
Waking up unrefreshed					

Latest Questionnaire Domain Scores (0-100)		
Sleep Quality & Architecture		91.7/100
Cognitive & Executive Function		83.3/100
Lab Biomarkers & Lifestyle		75/100

13

Digital Twin Risk Dashboard

Overall clinical risk: Moderate

Future Seizure Risk Moderate	Status Epilepticus Risk Low To Moderate	Hospitalization Risk Low To Moderate	Cognitive Decline Risk Low
Medication Toxicity Risk Low To Moderate	Functional Decline Risk Low	EEG-Derived Seizure Risk Elevated	Genetic Safety Flags 10 Flags, 7 High

State Vector (Health Metric Scores, 0-100)		
Disease Stability		67/100
Treatment Response		64/100
Medication Safety		67/100
Cognitive Health		69/100
Sleep Health		66/100
Quality Of Life		67/100
Each domain score is the mean of clinician-documented per-visit ratings across the clinical-note window; overall health score is the mean of the other domains. Derived digital-twin summary metrics, not a direct lab/imaging measurement.		

14 Clinical Action Plan

Priority Actions (Pharmacogenomics)

- HIGH** **Avoid or use with heightened monitoring**
See priority_safety_flags for the specific gene-drug pairs (ABCB1 (CC), BDNF (TC), CREB1 (AA), FKBP5 (TC), GRIA3 (A), HTR1B (TA), METTL21A (AA)).
- MODERATE** **Dose with caution / consider therapeutic drug monitoring**
See metabolizer_profile for reduced-function genes (CYP2C9, CYP2B6, CYP3A5, UGT1A4, POR).
- INFORMATIONAL** **Contextualize**
Interpret alongside actual clinical history, current medication list, and labs -- this panel alone does not establish a diagnosis.

Recommended Actions

- continue the documented antiseizure medication regimen unless changed by the treating clinician
- maintain the documented plan for consistent sleep schedule and trigger avoidance
- continue seizure and aura tracking across follow-up windows
- continue periodic CBC, liver function, and antiseizure drug-level safety monitoring when scheduled
- reassess at the planned follow-up interval

Monitoring Priorities

- seizure frequency by type
- seizure clustering
- average and longest seizure duration
- post-ictal recovery time
- aura pattern and severity
- sleep duration and sleep fragmentation
- missed medication doses and trigger exposure
- dizziness and other medication side effects
- CBC, liver function, and antiseizure drug-level tracking
- word-finding, memory, and attention symptoms
- Signal Variance
- Amplitude Dynamics
- Electrical Stability
- Preictal Detection
- Seizure Frequency
- Temporal Disease Progression

MRI Follow-Up

- Perform formal radiology review of the MRI image volumes.
- Run automated brain segmentation and longitudinal registration across the three study identifiers.
- Perform hippocampal volumetry because epilepsy-related mesial temporal abnormalities cannot be excluded from metadata alone.
- Run FLAIR white matter lesion analysis if clinically relevant.
- Compare future MRI studies using the same T1w and FLAIR acquisition strategy when possible.
- Correlate MRI findings with seizure semiology and neurological examination.
- Review MRI together with EEG findings for seizure-focus localization.
- Repeat or escalate imaging review if new neurological deficits, worsening seizures, or new focal symptoms occur.

Medication Reconciliation

- Current regimen reconciled and checked against curated reference pharmacology and this patient's own pharmacogenomic findings (Section 07). Predicted-model (SuperCYPsPred) coverage is still unresolved, so a live, comprehensive DDI screen is recommended before adding or changing any interacting medication.